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PORTABLE AIR CONDITIONER DEVICE

Abstract

A portable air conditioner device includes an ice chest for containing ice placed within the ice chest. A coiled tube is positioned within the interior space of the ice chest and the coiled tube is comprised of a thermally conductive material such that the coiled tube can be cooled by the ice. Additionally, the coiled tube has an outlet that extends through the ice chest. A blower is attached to the ice chest and the blower blows air into the coiled tube when the blower is turned on. In this way the coiled tube can cool the air blown into the coiled tube when the ice chest is filled with the ice thereby facilitating the cooled air to escape the outlet of the coiled tube. A condensation drain is attached to the coiled tube to receive condensed water from the air blown into the coiled tube.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] Not Applicable

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0002] Not Applicable

THE NAMES OF THE PARTIES TO A JOINT RESEARCH AGREEMENT

[0003] Not Applicable

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ON A COMPACT DISC OR AS A TEXT FILE VIA THE OFFICE ELECTRONIC FILING SYSTEM

[0004] Not Applicable

STATEMENT REGARDING PRIOR DISCLOSURES BY THE INVENTOR OR JOINT INVENTOR

[0005] Not Applicable

BACKGROUND OF THE INVENTION

(1) Field of the Invention

[0006] The disclosure relates to air conditioner devices and more particularly pertains to a new air conditioner device for producing cooled air in any location. The device includes an ice chest and a coiled tube that is positioned in the ice chest and the ice chest is filled with ice to cool the coiled tube. The device includes a blower that is attached to the ice chest which blows air into the coiled tube and outwardly through an output of the coiled tube that extends through the ice chest. The device includes a condensation drain that is in communication with the coiled tube to collect condensed water in the air that is blown through the coiled tube.

(2) Description of Related Art including information disclosed under 37 CFR 1.97 and 1.98

[0007] The prior art relates to air conditioner devices including a variety of portable air coolers that each at least comprises a bucket that can be filled with ice and a blower attached to the bucket for blowing air through the bucket to be cooled by the ice and a portable air cooler that comprises an ice chest and a fluid filled radiator disposed in the ice chest for cooling air that is blown through the ice chest by a blower. In no instance does the prior art disclose a portable air cooler that includes an ice chest and a coiled tube positioned in the ice chest that is cooled with ice placed in the ice chest and a blower which blows air through the coiled tube which is subsequently cooled by the coiled tube.

BRIEF SUMMARY OF THE INVENTION

[0008] An embodiment of the disclosure meets the needs presented above by generally comprising an ice chest for containing ice placed within the ice chest. A coiled tube is positioned within the interior space of the ice chest and the coiled tube is comprised of a thermally conductive material such that the coiled tube can be cooled by the ice. Additionally, the coiled tube has an outlet that extends through the ice chest. A blower is attached to the ice chest and the blower blows air into the coiled tube when the blower is turned on. In this way the coiled tube can cool the air blown into the coiled tube when the ice chest is filled with the ice thereby facilitating the cooled air to escape the outlet of the coiled tube. A condensation drain is attached to the coiled tube to receive condensed water from the air blown into the coiled tube.

[0009] There has thus been outlined, rather broadly, the more important features of the disclosure in order that the detailed description thereof that follows may be better understood, and in order that the present contribution to the art may be better appreciated. There are additional features of the disclosure that will be described hereinafter and which will form the subject matter of the claims appended hereto.

[0010] The objects of the disclosure, along with the various features of novelty which characterize the disclosure, are pointed out with particularity in the claims annexed to and forming a part of this disclosure.

Description

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWING(S)

[0011] The disclosure will be better understood and objects other than those set forth above will become apparent when consideration is given to the following detailed description thereof. Such description makes reference to the annexed drawings wherein:

[0012] FIG. **1** is a phantom perspective view of a portable air conditioner device according to an embodiment of the disclosure.

[0013] FIG. **2** is a front perspective view of an embodiment of the disclosure.

[0014] FIG. **3** is a top perspective view of an embodiment of the disclosure.

[0015] FIG. **4** is a top view of an embodiment of the disclosure.

[0016] FIG. **5** is a cross sectional view taken along line 5-5 of FIG. **4** of an embodiment of the disclosure.

[0017] FIG. **6** is a magnified detail view taken from circle **6** of FIG. **5** of an embodiment of the disclosure.

[0018] FIG. **7** is a bottom perspective phantom view of an embodiment of the disclosure.

[0019] FIG. **8** is a phantom in-use view of an embodiment of the disclosure.

DETAILED DESCRIPTION OF THE INVENTION

[0020] With reference now to the drawings, and in particular to FIGS. **1** through **8** thereof, a new air conditioner device embodying the principles and concepts of an embodiment of the disclosure and generally designated by the reference numeral **10** will be described.

[0021] As best illustrated in FIGS. **1** through **8**, the portable air conditioner device **10** generally comprises an ice chest **12** which defines an interior space **14** within the ice chest **12** to contain ice **16** placed within the ice chest **12**. Additionally, the ice chest **12** is comprised of a thermally insulating material to inhibit thermal communication between the ice **16** and ambient air. The ice chest **12** has a bottom wall **18** and a perimeter wall **20** extending upwardly from the bottom wall **18**. The perimeter wall **20** has an upper edge **22** which defines an opening **24** into the interior space **14** and the perimeter wall **20** has a pair of wheel wells **26** each recessed into a respective one of a first lateral side **28** and a second lateral side **30** of the perimeter wall **20**. Each of the pair of wheel wells **26** is aligned with an intersection with a back side **32** of the perimeter wall **20**. Each of the pair of wheel wells **26** has an upper bounding surface **34** that is concavely arcuate with respect to the bottom wall **18**.

[0022] The ice chest **12** has prominence **36** extending rearwardly from the back side **32** of the perimeter wall **20**. The prominence **36** is oriented to extend partially between the first lateral side **28** and the second lateral side **30** of the perimeter wall **20**. Additionally, the prominence **36** is positioned closer to the upper edge **22** of the perimeter wall **20** than the bottom wall **18** of the ice chest **12** and the prominence **36** has an upper surface **38** that is horizontally oriented. A pair of rollers **40** is each rotatably attached to the ice chest **12**. Each of the pair of rollers **40** is positioned in a respective one of the pair of wheel wells **26** to roll along a support surface **42** for transporting the ice chest **12**.

[0023] A lid **39** is hingedly coupled to the ice chest **12** for opening or closing the interior space **14** in the ice chest **12**. The lid **39** is comprised of a thermally insulating material to inhibit thermal communication between the ice **16** and ambient air when the lid **39** is closed. The lid **39** has a rear edge **41** that is hingedly coupled to the upper edge **22** of the perimeter wall **20** of the ice chest **12** corresponding to the back side **32** of the perimeter wall **20**. A handle **44** is provided which comprises a pair of telescopic members **46** and a grip **48** extending between the pair of telescopic members **46**. A lower end **50** of each of the pair of telescopic members **46** is attached to the upper surface **38** of the prominence **36**. Each of the pair of telescopic members **46** is urgeable into an extended condition having the grip **48** being spaced upwardly from the lid **39**. In this way the grip

48 can be gripped by a user **52** thereby facilitating the user **52** to roll the ice chest **12** along the support surface **42** for transporting the ice chest **12**.

[0024] A coiled tube **54** is positioned within the interior space **14** of the ice chest **12** and the coiled tube **54** defines a coil **55** that is vertically oriented on the ice chest **12**. The coiled tube **54** is comprised of a thermally conductive material, including but not being limited to steel or copper, such that the coiled tube **54** is cooled by the ice **16** when the interior space **14** is filled with the ice **16**. The coiled tube **54** has an inlet tube **56** and an outlet tube **58**; the inlet tube **56** is perpendicularly oriented with a top end **60** of the coil **55** defined by the coiled tube **54**.

Furthermore, the inlet tube **56** extends through the back side **32** of the perimeter wall **20** of the ice chest **12** at a point that is positioned closer to the upper edge **22** of the perimeter wall **20** than the bottom wall **18** of the ice chest **12**.

[0025] The outlet tube **58** has a lower portion **62** that is perpendicularly oriented with a bottom end **64** of the coil **55** defined by the coiled tube **54** and the lower portion **62** is vertically oriented in the interior space **14** of the ice chest **12**. Additionally, the lower portion **62** is surrounded by the coil **55** defined by the coiled tube **54**. The outlet tube **58** has an upper portion **66** that is perpendicularly oriented with the lower portion **62** and the upper portion **66** extends through a front side **68** of the perimeter wall **20** of the ice chest **12**.

[0026] A blower **70** is attached to the ice chest **12** and the blower **70** is in communication with the coiled tube **54**. The blower **70** blows air into the coiled tube **54** when the blower **70** is turned on to cool the air blown into the coiled tube **54** when the interior space **14** is filled with the ice **16**. The blower **70** is disposed on the back side **32** of the perimeter wall **20** of the ice chest **12**. Additionally, the blower **70** has an output **72** that is fluidly coupled to the inlet tube **56** of the coiled tube **54**. The blower **70** may comprise an electric air blower or the like and the blower **70** includes a battery **72** and a power button **73** for actuating or de-actuating the blower **70**.

[0027] A condensation drain **74** is provided and the condensation drain **74** is attached to the coiled tube **54**. The condensation drain **74** is in communication with the coil **55** defined by the coiled tube **54** to receive condensed water **76** from the air blown into the coiled tube **54**. The condensation drain **74** has an intake tube **78** that is in fluid communication with an intersection between the lower portion **62** of the outlet tube **58** of the coiled tube **54** and the bottom end **64** of the coiled tube **54**. Furthermore, as is most clearly shown in FIG. 5, the intake tube **78** extends toward the bottom wall **18** of the ice chest **12** to facilitate the condensed water **76** to travel downwardly in the intake tube **78** when the cooled air exits the bottom end **64** of the coiled tube **54** thereby separating the condensed water **76** from the cooled air.

[0028] The condensation drain **74** has an exhaust tube **80** which is fluidly coupled to the intake tube **78** to receive the condensed water **76** from the intake tube **78**. The exhaust tube **80** extends along the bottom wall **18** of the ice chest **12** and extends through the front side **68** of the perimeter wall **20** of the ice chest **12**. The exhaust tube **80** has a distal end **82** with respect to the front side **68** of the perimeter wall **20** and the distal end **82** is closed. Additionally, the exhaust tube **80** has a drain hole **84** which extends into an interior of the exhaust tube **80** and the drain hole **84** faces downwardly to drain the condensed water **76** from the condensation drain **74**. As is most clearly shown in FIG. 6, a cap **86** may be applied to the distal end **82** of the exhaust tube **80** to close the distal end **82** and the cap **86** may have a hole **88** that is aligned with the drain hole **84**.

[0029] In use, the ice chest **12** is filled with ice **16** and the lid **39** is closed and the blower **70** is turned on to blow air into the coiled tube **54**. The ice **16** in the ice chest **12** cools the coiled tube **54** which facilitates the coiled tube **54** to cool the air blown into the coiled tube **54**. The condensation drain **74** collects condensed water **76** from the cooled air to subsequently drain out of the drain hole in the exhaust tube **80**. The ice chest **12** can be positioned in any location that does not have air conditioning, such as a camper, a garage or any other enclosed space, thereby facilitating the cooled air to exit the outlet tube **58** of the coiled tube **54** to cool the enclosed space. The ice **16** can be replenished at any time when the ice **16** begins to melt and the battery in the blower **70** can be

replaced when the battery becomes depleted. In this way the enclosed space can be cooled for an extended duration of time.

[0030] With respect to the above description then, it is to be realized that the optimum dimensional relationships for the parts of an embodiment enabled by the disclosure, to include variations in size, materials, shape, form, function and manner of operation, device and use, are deemed readily apparent and obvious to one skilled in the art, and all equivalent relationships to those illustrated in the drawings and described in the specification are intended to be encompassed by an embodiment of the disclosure.

[0031] Therefore, the foregoing is considered as illustrative only of the principles of the disclosure. Further, since numerous modifications and changes will readily occur to those skilled in the art, it is not desired to limit the disclosure to the exact construction and operation shown and described, and accordingly, all suitable modifications and equivalents may be resorted to, falling within the scope of the disclosure. In this patent document, the word “comprising” is used in its non-limiting sense to mean that items following the word are included, but items not specifically mentioned are not excluded. A reference to an element by the indefinite article “a” does not exclude the possibility that more than one of the element is present, unless the context clearly requires that there be only one of the elements.

Claims

1. A portable air conditioner device for producing cooled air in any location, said device comprising: an ice chest defining an interior space within said ice chest wherein said interior space is configured to contain ice placed within said ice chest, said ice chest being comprised of a thermally insulating material wherein said ice chest is configured to inhibit thermal communication between the ice and ambient air; a pair of rollers, each of said pair of rollers being rotatably attached to said ice chest wherein each of said pair of rollers is configured to roll along a support surface for transporting said ice chest; a lid being hingedly coupled to said ice chest for opening or closing said interior space in said ice chest, said lid being comprised of a thermally insulating material wherein said lid is configured to inhibit thermal communication between the ice and ambient air when said lid is closed; a coiled tube being positioned within said interior space of said ice chest, said coiled tube defining a coil being vertically oriented on said ice chest, said coiled tube being comprised of a thermally conductive material wherein said coiled tube is configured to be cooled by the ice when said interior space is filled with the ice, said coiled tube having an outlet extending outwardly from said ice chest; a blower being attached to said ice chest, said blower being in communication with said coiled tube, said blower blowing air into said coiled tube when said blower is turned on wherein said coiled tube is configured to cool the air blown into said coiled tube when said interior space is filled with the ice thereby facilitating the cooled air to be blown outwardly through said outlet of said coiled tube; and a condensation drain being attached to said coiled tube, said condensation drain being in communication with said coil defined by said coiled tube wherein said condensation drain is configured to receive condensed water from the air blown into said coiled tube.

2. The device according to claim 1, wherein: said ice chest has a bottom wall and a perimeter wall extending upwardly from said bottom wall; said perimeter wall has an upper edge defining an opening into said interior space; said perimeter wall has a pair of wheel wells each being recessed into a respective one of a first lateral side and a second lateral side of said perimeter wall; each of said pair of wheel wells is aligned with an intersection with a back side of said perimeter wall; each of said pair of wheel wells has an upper bounding surface being concavely arcuate with respect to said bottom wall; said ice chest has a prominence extending rearwardly from said back side of said perimeter wall; said prominence is oriented to extend partially between said first lateral side and said second lateral side of said perimeter wall; said prominence is positioned closer to said upper

edge of said perimeter wall than said bottom wall of said ice chest; said prominence has an upper surface being horizontally oriented; each of said pair of rollers is positioned in a respective one of said pair of wheel wells; said device includes a handle comprising a pair of telescopic members and a grip extending between said pair of telescopic members; a lower end of each of said pair of telescopic members being attached to said upper surface of said prominence; each of said pair of telescopic members is urgeable into an extended condition having said grip being spaced upwardly from said lid wherein said grip is configured to be gripped by a user thereby facilitating the user to roll said ice chest along the support surface for transporting said ice chest; said lid has a rear edge being hingedly coupled to said upper edge of said perimeter wall of said ice chest corresponding to said back side of said perimeter wall.

3. The device according to claim 2, wherein: said coiled tube has an inlet tube and an outlet tube; said inlet tube is perpendicularly oriented with a top end of said coil defined by said coiled tube; said inlet tube extending through said back side of said perimeter wall of said ice chest at a point being positioned closer to said upper edge of said perimeter wall than said bottom wall of said ice chest; said outlet tube has a lower portion being perpendicularly oriented with a bottom end of said coil defined by said coiled tube having said lower portion being vertically oriented in said interior space of said ice chest; said lower portion is surrounded by said coil defined by said coiled tube; said outlet tube has an upper portion being perpendicularly oriented with said lower portion; and said upper portion extends through a front side of said perimeter wall of said ice chest such that said upper portion defines said outlet of said coiled tube.

4. The device according to claim 3, wherein: said blower is disposed on said back side of said perimeter wall of said ice chest; and said blower has an output being fluidly coupled to said inlet tube of said coiled tube.

5. The device according to claim 3, wherein said condensation drain has an intake tube being in fluid communication with an intersection between said lower portion of said outlet tube of said coiled tube and said bottom end of said coiled tube having said intake tube extending toward said bottom wall of said ice chest wherein said intake tube is configured to facilitate the condensed water to travel downwardly in said intake tube when the cooled air exits said bottom end of said coiled tube thereby separating the condensed water from the cooled air.

6. The device according to claim 5, wherein: said condensation drain has an exhaust tube being fluidly coupled to said intake tube wherein said exhaust tube is configured to receive the condensed water from said intake tube; said exhaust tube extends along said bottom wall of said ice chest and extends through said front side of said perimeter wall of said ice chest; said exhaust tube has a distal end with respect to said front side of said perimeter wall; said distal end is closed; said exhaust tube has a drain hole extending into an interior of said exhaust tube; and said drain hole faces downwardly wherein said drain hole is configured to drain the condensed water from said condensation drain.

7. A portable air conditioner device for producing cooled air in any location, said device comprising: an ice chest defining an interior space within said ice chest wherein said interior space is configured to contain ice placed within said ice chest, said ice chest being comprised of a thermally insulating material wherein said ice chest is configured to inhibit thermal communication between the ice and ambient air, said ice chest having a bottom wall and a perimeter wall extending upwardly from said bottom wall, said perimeter wall having an upper edge defining an opening into said interior space, said perimeter wall having a pair of wheel wells each being recessed into a respective one of a first lateral side and a second lateral side of said perimeter wall, each of said pair of wheel wells being aligned with an intersection with a back side of said perimeter wall, each of said pair of wheel wells having an upper bounding surface being concavely arcuate with respect to said bottom wall, said ice chest having prominence extending rearwardly from said back side of said perimeter wall, said prominence being oriented to extend partially between said first lateral side and said second lateral side of said perimeter wall, said prominence being positioned closer to

said upper edge of said perimeter wall than said bottom wall of said ice chest, said prominence having an upper surface being horizontally oriented; a pair of rollers, each of said pair of rollers being rotatably attached to said ice chest, each of said pair of rollers being positioned in a respective one of said pair of wheel wells wherein each of said pair of rollers is configured to roll along a support surface for transporting said ice chest; a lid being hingedly coupled to said ice chest for opening or closing said interior space in said ice chest, said lid being comprised of a thermally insulating material wherein said lid is configured to inhibit thermal communication between the ice and ambient air when said lid is closed, said lid having a rear edge being hingedly coupled to said upper edge of said perimeter wall of said ice chest corresponding to said back side of said perimeter wall; a handle comprising a pair of telescopic members and a grip extending between said pair of telescopic members, a lower end of each of said pair of telescopic members being attached to said upper surface of said prominence, each of said pair of telescopic members being urgeable into an extended condition having said grip being spaced upwardly from said lid wherein said grip is configured to be gripped by a user thereby facilitating the user to roll said ice chest along the support surface for transporting said ice chest; a coiled tube being positioned within said interior space of said ice chest, said coiled tube defining a coil being vertically oriented on said ice chest, said coiled tube being comprised of a thermally conductive material wherein said coiled tube is configured to be cooled by the ice when said interior space is filled with the ice, said coiled tube having an outlet extending outwardly from said ice chest, said coiled tube having an inlet tube and an outlet tube, said inlet tube being perpendicularly oriented with a top end of said coil defined by said coiled tube, said inlet tube extending through said back side of said perimeter wall of said ice chest at a point being positioned closer to said upper edge of said perimeter wall than said bottom wall of said ice chest, said outlet tube having a lower portion being perpendicularly oriented with a bottom end of said coil defined by said coiled tube having said lower portion being vertically oriented in said interior space of said ice chest, said lower portion being surrounded by said coil defined by said coiled tube, said outlet tube having an upper portion being perpendicularly oriented with said lower portion, said upper portion extending through a front side of said perimeter wall of said ice chest such that said upper portion defines said outlet of said coiled tube; a blower being attached to said ice chest, said blower being in communication with said coiled tube, said blower blowing air into said coiled tube when said blower is turned on wherein said coiled tube is configured to cool the air blown into said coiled tube when said interior space is filled with the ice, said blower being disposed on said back side of said perimeter wall of said ice chest, said blower having an output being fluidly coupled to said inlet tube of said coiled tube; and a condensation drain being attached to said coiled tube, said condensation drain being in communication with said coil defined by said coiled tube wherein said condensation drain is configured to receive condensed water from the air blown into said coiled tube, said condensation drain having an intake tube being in fluid communication with an intersection between said lower portion of said outlet tube of said coiled tube and said bottom end of said coiled tube having said intake tube extending toward said bottom wall of said ice chest wherein said intake tube is configured to facilitate the condensed water to travel downwardly in said intake tube when the cooled air exits said bottom end of said coiled tube thereby separating the condensed water from the cooled air, said condensation drain having an exhaust tube being fluidly coupled to said intake tube wherein said exhaust tube is configured to receive the condensed water from said intake tube, said exhaust tube extending along said bottom wall of said ice chest and extending through said front side of said perimeter wall of said ice chest, said exhaust tube having a distal end with respect to said front side of said perimeter wall, said distal end being closed, said exhaust tube having a drain hole extending into an interior of said exhaust tube, said drain hole facing downwardly wherein said drain hole is configured to drain the condensed water from said condensation drain.
